

1 Numbers and the number system

	Students should be taught to:	Notes
1.1 Integers	A understand and use integers (positive, negative and zero)	
	B understand place value	
	C use directed numbers in practical situations	e.g. temperatures
	D order integers	
	E use the four rules of addition, subtraction, multiplication and division	
	F use brackets and the hierarchy of operations	
	G use the terms 'odd', 'even', 'prime numbers', 'factors' and 'multiples'	
	H identify prime factors, common factors and common multiples	
1.2 Fractions	A understand and use equivalent fractions, simplifying a fraction by cancelling common factors	$\frac{8}{60} = \frac{2}{15}$ in its simplest form (lowest terms)
	B understand and use mixed numbers and vulgar fractions	
	C identify common denominators	
	D order fractions and calculate a given fraction of a given quantity	
	E express a given number as a fraction of another number	
	F use common denominators to add and subtract fractions and mixed numbers	$\frac{2}{3} + \frac{5}{7}, \quad 3\frac{1}{5} - 2\frac{2}{3}$
	G convert a fraction to a decimal or a percentage	$\frac{3}{5} = 0.6 = 60\%$ $\frac{4}{9} = 0.4444... = 44.4...\%$
	H understand and use unit fractions as multiplicative inverses	$3 \div 5 = 3 \times \frac{1}{5}$

	Students should be taught to:	Notes
1.3 Decimals	A use decimal notation	
	B understand place value	
	C order decimals	
	D convert a decimal to a fraction or a percentage	Terminating decimals only
	E recognise that a terminating decimal is a fraction	$0.65 = \frac{65}{100} = \frac{13}{20}$
1.4 Powers and roots	A identify square numbers and cube numbers	
	B calculate squares, square roots, cubes and cube roots	
	C use index notation and index laws for multiplication and division of positive and negative integer powers including zero	
	D express integers as a product of powers of prime factors	$720 = 2^4 \times 3^2 \times 5$
	E find highest common factors (HCF) and lowest common multiples (LCM)	
1.5 Set language and notation	A understand the definition of a set	
	B use the set notation $\cup$, $\cap$ and $\in$ and $\notin$	$\mathcal{U}$ = universal set $\emptyset$ = empty set
	C understand the concept of the universal set and the empty set and the symbols for these sets	
	D understand and use the complement of a set	Use the notation A'
	E use Venn diagrams to represent sets	

	Students should be taught to:	Notes
1.6 Percentages	A understand that 'percentage' means 'number of parts per 100'	
	B express a given number as a percentage of another number	
	C express a percentage as a fraction and as a decimal	
	D understand the multiplicative nature of percentages as operators	$15\% \text{ of } 120 = \frac{15}{100} \times 120$
	E solve simple percentage problems, including percentage increase and decrease	
	F use reverse percentages	In a sale, prices were reduced by 30%. The sale price of an item was £17.50 Calculate the original price of the item
	G use compound interest and depreciation	
1.7 Ratio and proportion	A use ratio notation, including reduction to its simplest form and its various links to fraction notation	Express in the form $1 : n$
	B divide a quantity in a given ratio or ratios	Share £416 in the ratio 5 : 3 or 4 : 3 : 1
	C use the process of proportionality to evaluate unknown quantities	
	D calculate an unknown quantity from quantities that vary in direct proportion	s varies directly as t Find the missing value in a table
	E solve word problems about ratio and proportion	Including maps and scale diagrams
1.8 Degree of accuracy	A round integers to a given power of 10	
	B round to a given number of significant figures or decimal places	
	C identify upper and lower bounds where values are given to a degree of accuracy	
	D use estimation to evaluate approximations to numerical calculations	By rounding values to 1 significant figure
1.9 Standard form	A calculate with and interpret numbers in the form $a \times 10^n$ where n is an integer and $1 \leq a < 10$	$150\,000\,000 = 1.5 \times 10^8$

	Students should be taught to:	Notes
1.10 Applying number	A use and apply number in everyday personal, domestic or community life	
	B carry out calculations using standard units of mass, length, area, volume and capacity	Metric units only
	C understand and carry out calculations using time, and carry out calculations using money, including converting between currencies	
1.11 Electronic calculators	A use a scientific electronic calculator to determine numerical results	